

ASSESSMENT OF THEMATIC MAPPING TECHNIQUES FOR POPULATION DISTRIBUTION VISUALIZATION: A Comparative study of Dot Density, Filled Isopleth and Unfilled Isopleth

Abstract

Thematic maps play a critical role in communicating spatial patterns and supporting geographic decision making. Among the various methods used to represent population density, Dot Density maps, Filled Isopleths (Colour-graded), and Unfilled Isopleths (Contour-based) are widely applied, yet their influence on users' perception and interpretation of spatial density patterns remains insufficiently understood. This study investigates how these three cartographic representations affect the perception of population density, with particular emphasis on hotspot identification and spatial gradient interpretation. A between-subjects online experiment was conducted using maps representing both real-world and synthetic population distributions. Participants completed tasks while measuring accuracy, response time, and perceived difficulty were collected. Demographic information and self-reported cartographic experience were also recorded.

The findings demonstrate that thematic map design substantially affects the interpretation of population density patterns. Filled Isopleths appear to provide the most effective support for both hotspot identification and gradient interpretation, while Dot Density maps remain useful for rapid exploratory assessment of spatial clustering. These results contribute to cartographic perception research and provide practical recommendations for selecting thematic map types in population mapping and related spatial analysis applications.

Keywords: Cartographic perception, Thematic Maps, Population Density, Filled isopleth, Unfilled Isopleth, Dot Density, Real-World, Synthetic, Hotspot Identification, Spatial Gradients, Map Usability, Spatial Cognition.

1.Introduction

Thematic maps are special types of maps that show information about a topic, such as population, income, or temperature. They are important in cartography to visualize spatial patterns and support analysis (Slocum et al., 2023).

The visualization of spatial phenomena plays a critical role in understanding geographic patterns and support decision making processes. Thematic maps are widely used to communicate complex spatial information effectively through appropriate symbolization and map design (Slocum et al., 2023). In the context of natural hazard assessment, thematic mapping techniques can be used to represent population distribution built-up areas, and hazard exposure. Ehrlich et al., (2018) demonstrated the importance of integrating population density and built-up area datasets derived from remote sensing to quantify exposure to natural hazards to multiple spatial scales. Several mapping techniques are commonly used to represent population distribution. Dot Density maps use repeated dots to represent quantities within geographic space, enabling users to visually identify clustering, concentration, and dispersion patterns. Filled Isopleth represents continuous spatial variation through graduated colour surfaces or heatmap-style visualization, allowing smooth transitions between density values. In contrast, Unfilled Isopleth, commonly referred to as isolines or contour maps, use lines connecting points of equal value to represent spatial gradients and continuous phenomena without filled colour regions. Unlike discrete spatial representations, isopleth-based visualization estimates values continuously across geographic space (Traun et al., 2017; He et al., 2011; Slocum et al., 2023). All three techniques are valid; each emphasizes different spatial characteristics and may lead to distinct interpretations by map readers. Previous studies have shown that thematic map design can influence user effectiveness, attractiveness ratings, and overall evaluation of spatial information. Furthermore, different visualization methods may affect how spatial patterns are perceived and interpreted, highlighting the importance of selecting an appropriate mapping technique for a given task. Selecting the most appropriate method therefore plays a crucial role in thematic design (Medynska-Gulij et al., 2020; Besançon et al., 2020; Forrest & Medynska-Gulij, 2021).

The effectiveness of these thematic mapping techniques is closely related to cartographic communication and spatial cognition, as different visualization methods requires user to interpret spatial information through varying levels of abstraction and visual complexity (MacEachren 2004). However, different thematic mapping techniques emphasize different

characteristics of spatial data. Because these visualization methods communicate information in different ways, they may influence how users perceive, interpret, and evaluate spatial patterns.

1.1 Map Interpretation and User-Centred Design

Map design influences how users read, understand, and interpret spatial information. Thematic maps use abstraction, classification, colour, symbols, and visual hierarchy to communicate complex geographic patterns in a simpler form.

From a user-centred cartographic perspective, the effectiveness of a map depends not only on the data being mapped, but also on the user, the visual design, and the task being performed. Previous studies have shown that different map designs can affect interpretation of accuracy, response time, perceived difficulty, and overall map-reading performance.

This study follows a user-centred approach by evaluating how different thematic map types of support users in identifying population hotspots and interpreting spatial gradients. The evaluation is based on three usability measures: accuracy, response time, and perceived difficulty.

1.2 Research Gap

Previous studies have shown that thematic map type can influence map-reading performance. However, many existing studies focus mainly on map types such as choropleth maps, graduated symbol maps, and isoline maps.

There is still limited user-based research comparing Dot Density maps, Filled Isopleths, and Unfilled Isopleths within the same experimental framework, especially for population density visualization.

This study addresses that gap by comparing these three map types using both real-world and synthetic population datasets. It evaluates their effectiveness for two common interpretation tasks: hotspot identification and spatial gradient interpretation.

1.3 Aim of the Study

The aim of this study is to assess how different thematic mapping techniques influence the interpretation of spatial population density patterns.

The study compares Dot Density maps, Filled Isopleths, and Unfilled Isopleths using real-world and synthetic datasets to understand how each representation affects users' accuracy, response time, and perceived difficulty.

1.4 Research Objectives

1.4.1 Main Objective

To assess the impact of different thematic mapping techniques on the interpretation of spatial population distribution patterns.

1.4.2 Specific Objectives

Compare the effectiveness of Dot Density maps, Filled Isopleths, and Unfilled Isopleths for hotspot identification.

- Evaluate how the three map types support spatial gradient interpretation.
- Compare interpretation accuracy and response time across the map types.
- Assess perceived difficulty associated with each map type and task.
- Investigate the relationship between perceived difficulty and actual task performance.
- Evaluate map usability using both synthetic and real-world population datasets.
- Provide recommendations for selecting appropriate thematic mapping techniques for population density visualization.

2. Methods

This study used a controlled web-based user experiment to evaluate how different thematic map representations influence the interpretation of population density patterns.

A between-subjects experimental design was used. Participants were randomly assigned to one of three map conditions:

- Dot Density maps (DD)
- Filled Isopleths (FI)
- Unfilled Isopleths (UI)

Each participant interacted with only one map type throughout the experiment. This helped reduce learning effects, visual familiarity, and fatigue.

The experiment focused on two spatial interpretation tasks:

- Hotspot identification
- Spatial gradient interpretation

Map performance was evaluated using three measures:

- Interpretation accuracy
- Response time
- Perceived task difficulty

Both real-world and synthetic datasets were used. The real-world datasets provided realistic geographic contexts, while synthetic datasets were generated in Python to create controlled hotspot and gradient patterns.

All maps were created from the same underlying datasets and were standardized using consistent layouts, scales, legends, and visual design settings. Dot Density maps represented population through point symbols, while Filled and Unfilled Isopleths were generated from continuous density surfaces.

Survey data were collected using LimeSurvey and analysed using statistical methods such as Chi-square tests, Kruskal-Wallis tests, and a Generalized Linear Mixed Model (GLMM).

Survey Link: <https://survey.plus.ac.at/index.php/624757?lang=en>

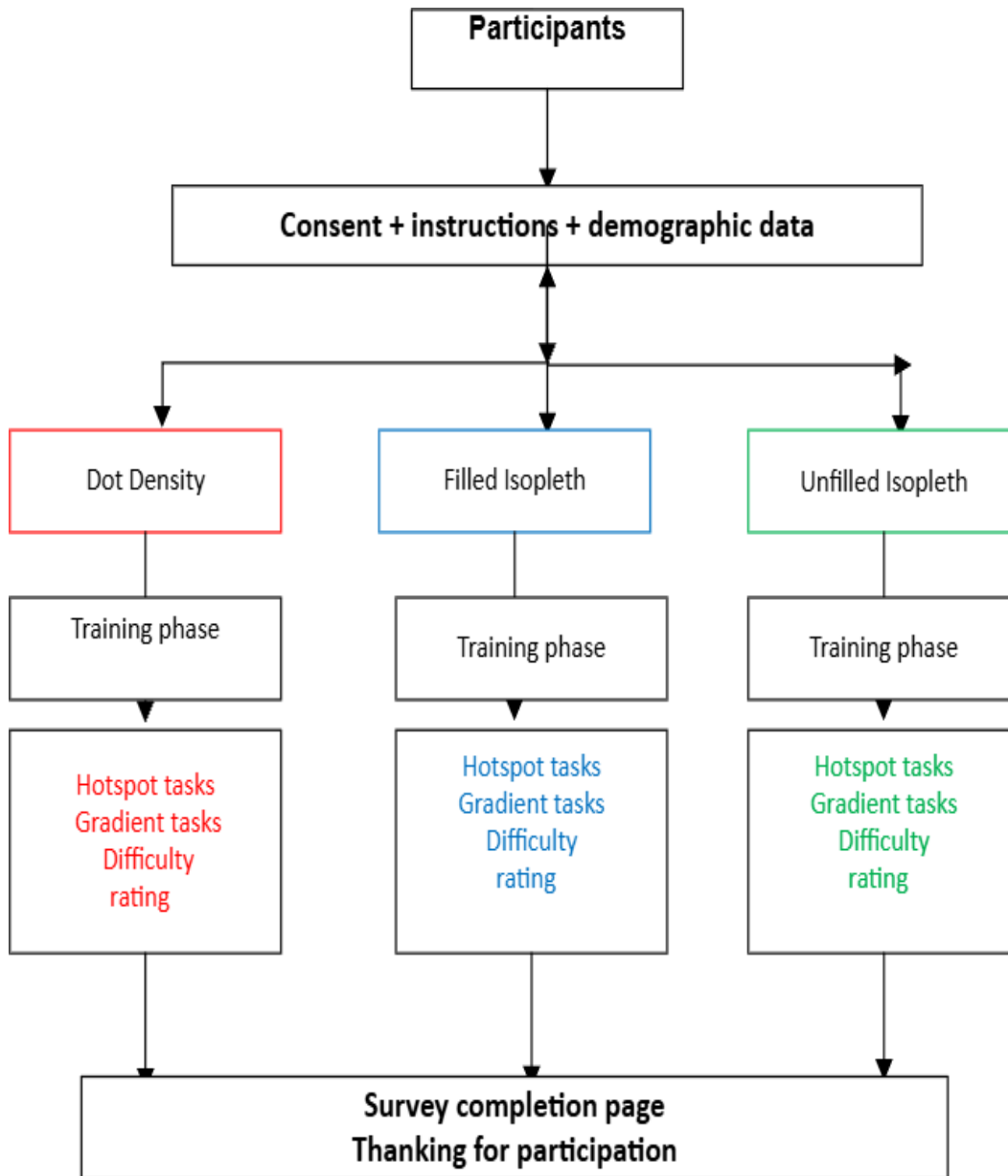


Figure 1: Graphical overview of the experimental workflow and random assignment procedure used on Limesurvey

2.1 Data Analysis

Data preprocessing and thematic map generation were carried out using GIS and spatial analysis tools, including ArcGIS Pro and Python.

The survey data were cleaned and organized in Excel before statistical analysis. The analysis focused on comparing the performance of the three thematic map types:

- Dot Density maps
- Filled Isopleths
- Unfilled Isopleths

Descriptive statistics were used to summarize participant information, interpretation of accuracy, response time, and perceived difficulty ratings.

Comparative statistical tests were then used to examine whether performance differed between the three map types. The analysis also investigated the relationship between perceived task difficulty and actual task performance.

Statistical analysis was conducted in Jamovi, an open-source statistical software built on R. Jamovi was selected because it provides reliable statistical tools through a user-friendly interface, making it suitable for academic quantitative analysis.

2.1.1 Statistical Methods Used

- Descriptive statistics
- Chi-square tests for accuracy comparisons
- Kruskal-Wallis tests for response time and difficulty comparisons
- Post-hoc tests where required
- Spearman correlation analysis
- Generalized Linear Mixed Model (GLMM)

3. Main Thesis Results

The results show that the choice of thematic map representation influenced how users interpreted population density patterns.

Overall, Filled Isopleths achieved the highest interpretation of accuracy and were generally perceived as the easiest map type to use. Dot Density maps often supported faster responses, while Unfilled Isopleths were associated with higher perceived difficulty and lower overall performance.

3.1 Overall Performance

The three map types were compared using:

- Interpretation accuracy
- Response time

- Perceived task difficulty

The overall results suggest that Filled Isopleths provided the strongest balance between accuracy and usability, while Dot Density maps were useful for quick visual interpretation.

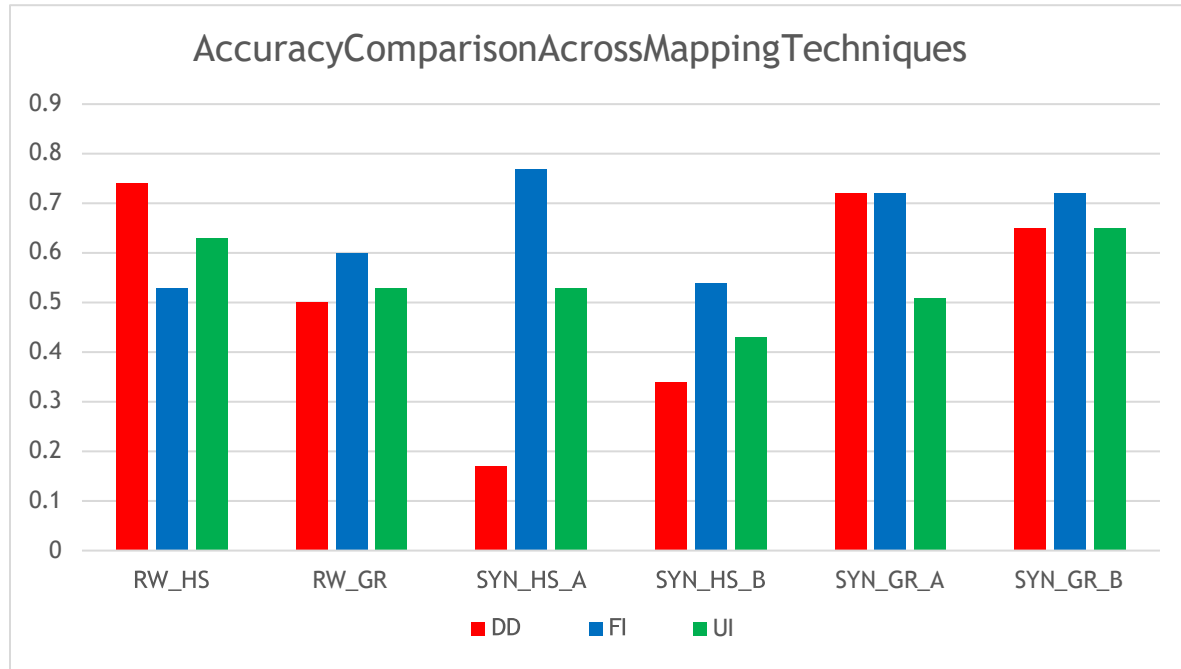


Figure 3: Average accuracy scores across thematic mapping techniques for real-world and synthetic spatial interpretation tasks.

Table 1. Inferential Statistical analysis of answer accuracy across the three thematic map types.

Task	Chi-Square Overall χ^2	P	Cramér's V	Pairwise Comparison	χ^2	Cramér's V	P	Significant
RW_HS	5.66	0.059	0.178	DD-FI	5.64	0.217	0.018	Yes
				DD-UI	1.45	0.112	0.229	-----
				FI-UI	1.38	0.107	0.241	-----
RW_GD	1.17	0.557	0.811	DD-FI	0.13	0.0972	0.136	-----
				DD-UI	0.138	0.0345	0.710	-----
				FI-UI	0.477	0.0628	0.491	-----
SYN_HS_A	43.8	< .001	0.496	DD-FI	43.5	0.602	< .001	Yes
				DD-UI	0.853	0.379	< .001	Yes
				FI-UI	3.09	0.253	0.006	Yes
SYN_HS_B	5.09	0.079	0.169	DD-FI	5.02	0.204	0.025	Yes
				DD-UI	0.908	0.0885	0.341	-----
				FI-UI	1.65	0.117	0.199	-----
SYN_GD_A	7.49	0.024	0.205	DD-FI	0.000	0.002	0.984	-----
				DD-UI	5.27	0.213	0.022	Yes
				FI-UI	5.56	0.215	0.018	Yes
SYN_GD_B	0.926	0.629	0.0712	DD-FI	0.701	0.0764	0.402	-----
				DD-UI	0.00	0.00	1.00	-----
				FI-UI	0.701	0.0764	0.402	-----

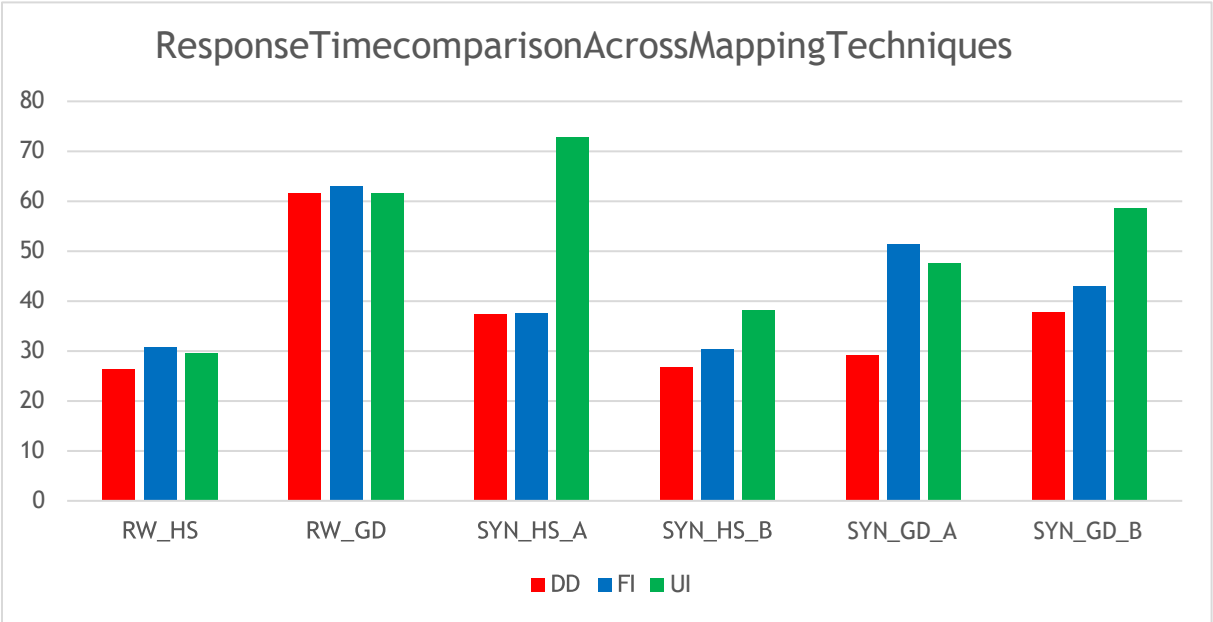


Figure 4: Average response time across thematic mapping techniques for real-world and synthetic spatial tasks

Table 2. Inferential Statistical analysis of response time across the three thematic map types.

Task	<i>Kruskal Wallis H</i>	<i>P</i>	Map Type	M (s)	SD	Post Hoc Groups	<i>p</i>
RW_HS	2.35	0.309	DD	26.3	15.8	-----	Not
			FI	30.72	17.9		Statistically
			UI	29.5	22.8		Significant
RW_GD	7.00	0.30	DD	61.5	46.0	DD-FI	1.000
			FI	63.0	39.3	DD-UI	0.165
			UI	61.5	81.3	FI-UI	0.33
SYN_GD_A	12.10	0.002	DD	29.2	14.6	DD-FI	0.012
			FI	51.3	77.9	DD-UI	0.005
			UI	47.5	33.7	FI-UI	1.000
SYN_GD_B	4.99	0.082	DD	37.7	25.8	-----	Not
			FI	43.0	21.2		Statistically
			UI	58.6	67,5		Significant
SYN_HS_A	6.67	0.36	DD	37.3	28.9	DD-FI	0.937
			FI	37.5	39.7	DD-UI	0.376
			UI	72.7	137	FI-UI	0.031
SYN_HS_B	2.16	0.340	DD	26.8	19.6		Not
			FI	30.4	18.3		statistically
			UI	38.2	41.7		Significant

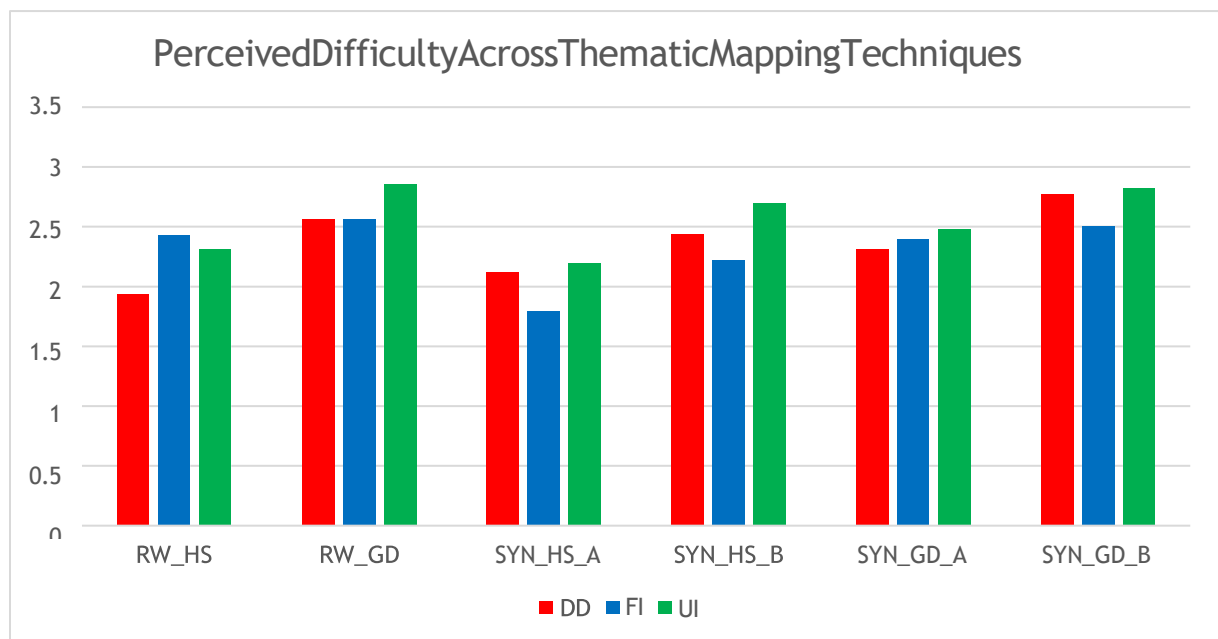


Figure 5: Average difficulty rating across thematic mapping techniques for real-world and synthetic spatial interpretation tasks

3.2 Hotspot and Gradient Interpretation

The findings also show that map effectiveness depends on the type of task being performed.

Filled Isopleths were particularly effective for representing continuous density patterns and supporting spatial gradient interpretation. Dot Density maps remained useful for identifying local clustering and hotspot patterns, but they were less consistent in overall accuracy.

This suggests that no single map type is universally best. Instead, the most suitable representation depends on the analytical task and the type of spatial pattern being interpreted.

Table 3. Inferential Statistical analysis of perceived difficulty rating across the three thematic map types.

Task	Kruskal-Wallis's X^2	p	Significant Pairwise Comparison	p
RW_HS	8.37	0.015	DD:FI	0.026
RW_GRD	3.05	0.218	-----	-----
SYN_HS_A	7.58	0.023	FI:UI	0.029
SYN_HS_B	7.20	0.027	FI:UI	0.022
SYN_GRD_A	0.79	0.674	-----	-----
SYN_GRD_B	4.25	0.119	-----	-----

3.3 Real-World and Synthetic Dataset Comparison

The comparison between real-world and synthetic datasets showed similar performance trends across different spatial contexts.

The real-world datasets provided realistic geographic complexity, while the synthetic datasets allowed controlled testing of hotspot and gradient patterns. Together, they helped confirm that the observed differences were mainly related to the map representation rather than only the dataset type.

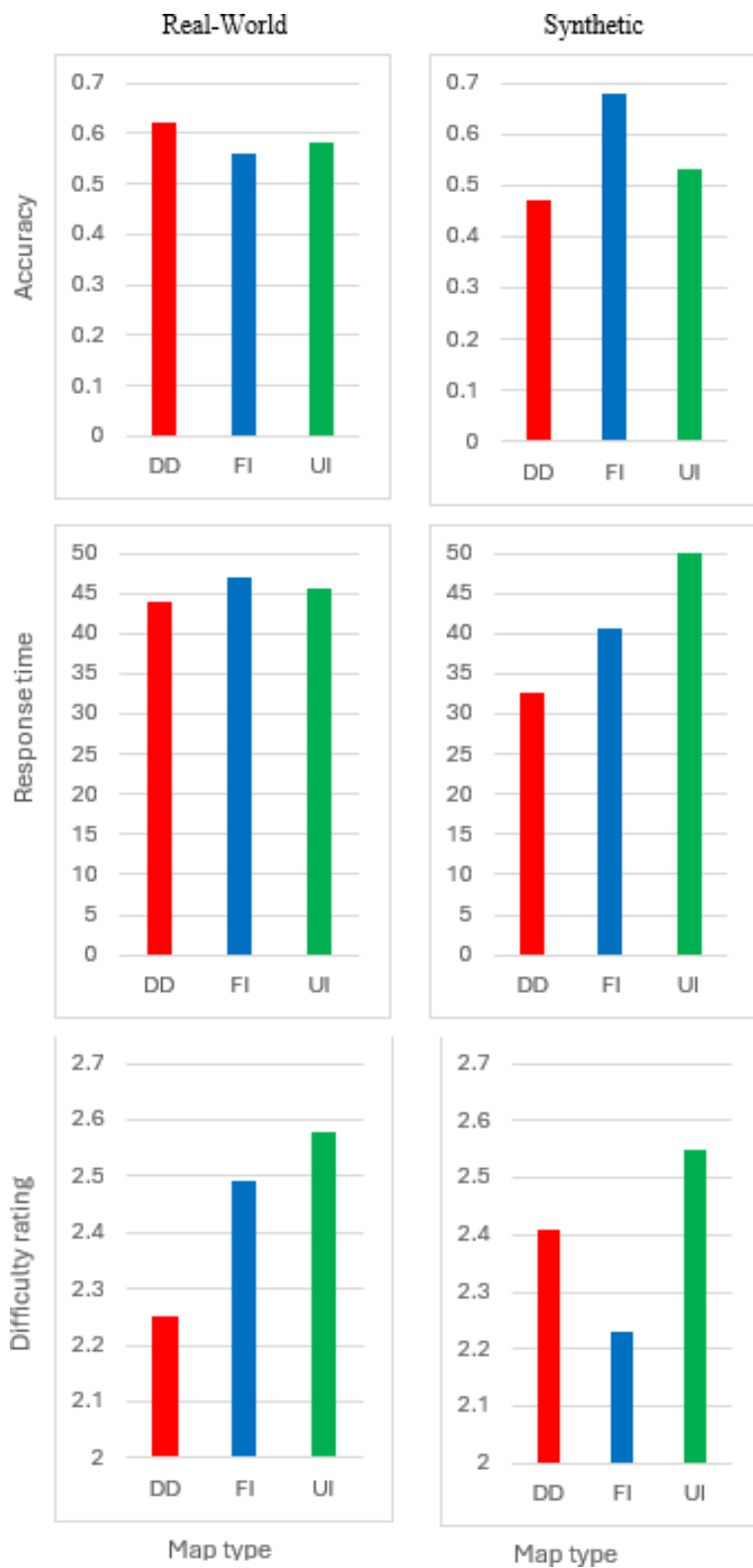


Figure 6: Comparison of interpretation performance between real-world and synthetic spatial tasks across thematic mapping techniques. Real-world tasks include RW_HS and RW_GD, while synthetic tasks include SYN_HS_A, SYN_HS_B, SYN_GD_A, and SYN_GD_B

3.4 Participant Characteristics

Participant characteristics such as age, gender, educational level, and cartographic experience had limited influence on overall task performance.

This suggests that the main performance differences were primarily driven by the map representations themselves rather than by demographic differences between participants.

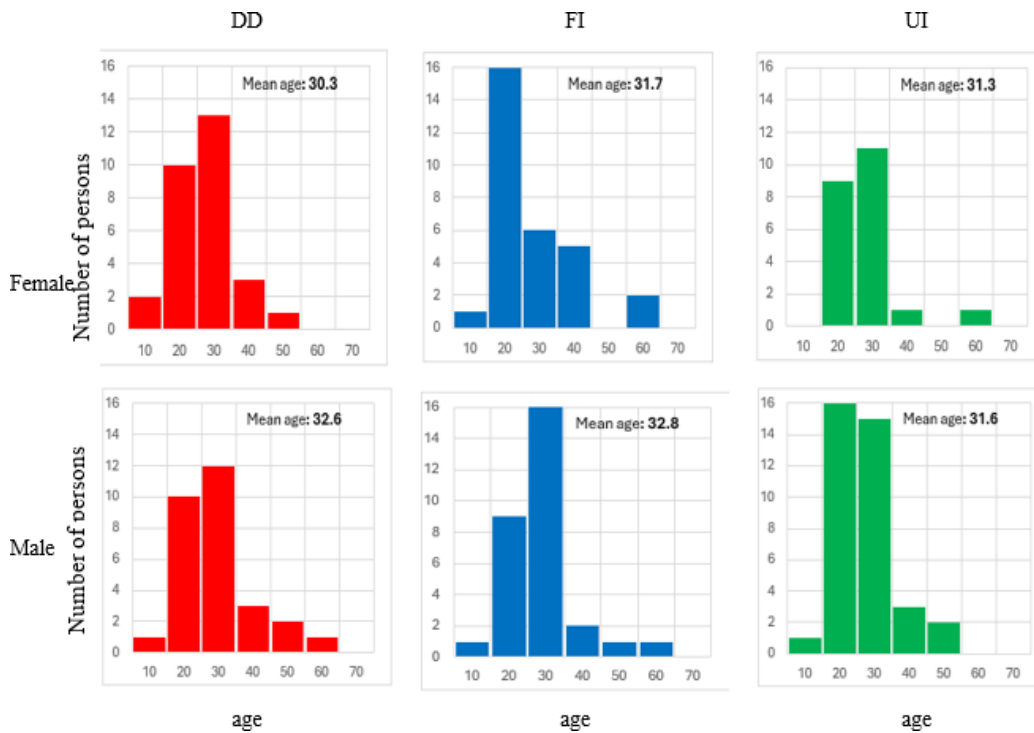


Figure 7: Age distribution and group

Table 1. Participants educational level (Raw count)

	DD	FI	UI	Σ
High School	9	10	4	23
Bachelor's Degree	27	31	30	88
Master's Degree	17	15	15	47
PhD / Doctorate	3	4	4	11
Others	2	2	5	9
Σ	58	62	58	178

Table 2: Self-reported cartographic experience (Raw count)

	DD	FI	UI	Σ
Beginner	15	7	9	31
Basic	5	5	5	15
Intermediate	15	23	11	49
Advanced	13	15	24	52
Expert	10	12	9	31
Σ	58	62	58	178

3.5 Summary of Findings

The main findings of the thesis are:

- Filled Isopleths achieved the highest overall accuracy.
- Dot Density maps support faster visual interpretation.
- Unfilled Isopleths were generally more difficult to interpret.
- Map effectiveness depended on the analytical task.
- Real-world and synthetic datasets showed consistent performance trends.
- Participant characteristics had limited influence on overall results.

Overall, the study highlights the importance of selecting thematic mapping techniques based on the intended task, the spatial pattern being visualized, and the target audience.

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